

Know Your Users: Towards Explainable AI in Bangladesh

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ABSTRACT

When we are talking about explainable AI and trying to come out of the black box by going beyond algorithmic transparency, we are being ignorant about a big user community. Although XAI research has advanced over time, there hasn't been much study done on the development, evaluation, and application of explainability methodologies in the global south. In this paper, we focus on Bangladesh, which is a part of the South, to understand the AI user community of this region and show how the explainability needs are different for different users and who should XAI focus on. Our work reflects on the unique needs and constraints of the region and recommends potential directions for accessible and human-centered explainability research. We argue that before developing technology and systems, human requirements should be assessed and comprehended.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*.

KEYWORDS

XAI for U; Explainable AI; user centric explanation; xai

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1 INTRODUCTION

In recent years, artificial intelligence (AI) has become the backbone of digital assistance in terms of everything and is accessible through cell phones easily. AI is the primary force behind developing technologies like generative AI, big data, robotics, and the Internet of Things (IoT). With open AI's recent achievements, including GPT-4, the biggest and most sophisticated AI model for language processing available, which has been applied to many different tasks such as chatbots, text summarization, and language translation, generative AI has reached a considerably higher level.

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Not only that, but AI is becoming more and more commonplace and assisting with important human decisions, including early disease detection of deadly disease cancer and stroke [7, 8]. Therefore, AI must be understandable to its end-users, but the machine's inability to explain its actions and reasoning for the decisions might limit the acceptability of these systems. Due to these difficulties, explainable AI (XAI) research is becoming more popular. XAI makes it possible for human users to understand the rationale behind the output produced by AI-driven machine learning (ML) algorithms.

Hence, the XAI system should be able to describe its capabilities and understandings, what it has done, what it is doing right now, and what will happen next [6]. The majority of XAI explanations still lack usability, practical interpretability, and effectiveness for actual users [1, 19, 33, 34]. The XAI end users are those who depend on decisions, recommendations, or actions made by an AI system [5]. Depending on the community context, there are numerous ways to evaluate and operationalize the XAI systems. To make XAI more explainable from a socio-technical perspective to a human-centered perspective, much research has been conducted to understand the user needs [3, 32]. Research addressing the sociotechnical challenges resulting from the usage of AI systems in the Global North has rapidly increased in recent years [20]. However, the majority of existing literature on XAI predominantly concentrates on research conducted within Western geography or context [9]. Consequently, there is a discernible scarcity of studies examining the application of XAI in other geographical regions. Very little or no research has been done to situate or ground explainability in the context of the south, especially in Bangladesh. In India, studies are designed to understand what explainability means for the underprivileged, marginalized, and unlettered population in rural areas [28]. Urge has also been seen in the work of Ehsan et al. [3] to identify the different 'whos' (relevant stakeholders and end users) to understand their needs since actionable explanations can inform the user to take better actions and help them to make the AI systems accountable.

Making explainability meaningful and accessible for all users, from developers to end users, in both esteemed and marginalized communities is still a challenge. In our work, we are focusing on the South, particularly Bangladesh. Bangladesh is the second largest South Asian country economically. Moreover, the nation's geographic position naturally positions it as a key component of the US, India, Japan, and Australia's Indo-Pacific strategy. The eighth-most populous nation in the world is Bangladesh, where 77.36 million people are using the internet and is a technology-emergent country [12]. All things considered, AI is widely used in Bangladesh. Some examples of this include ride-sharing, chatbots, natural language processing (NLP) for Bengali, hotel and ticket booking, and

real-time mapping. Since the country is gradually moving onto AI era there is a great opportunity to use explainable AI to ensure fair usage of technology. We aim to enlighten the XAI community about the end users and their explainability needs in this region to help the developers re-imagine the explainable methods in the South to make user-centered explanations for AI-driven systems.

2 XAI PERSPECTIVE FROM SOUTH

XAI aims to provide explanations to the end users of AI products who are dependent on the system's actions, decisions, and recommendations. There are various kinds of users, from the development phase to uses of the system [13]. An expert - who have enough domain-specific knowledge; critic is someone who expresses evaluations and opinions about different kinds of creative works, including AI systems, music, film, and art or operator those who are adept at using AI techniques to produce useful outcomes, intelligence analysts, - who have strong ability to solve problems, and civilian society including labour unions, religious organisations, indigenous groups, humanitarian organisations, non-governmental organisations (NGOs) are examples of different types of users. They also have variations according to their educational background and profession. On the other hand, a developer who build, design, and deploy AI systems and models or test operator who checks the system's functionality needs to know where there might be room for improvement may be one of the users who requests an explanation of the system. Policymakers, whose responsibility is to further serve the needs of a wide range of stakeholders might attempt to gauge the system's fairness, could be yet another user group. Each user group might have a different demand for the explanatory style that they find to be the most efficient to understand. The end users (local people) might need different types of explanations according to their needs, context, culture, education, and digital literacy level.

While the XAI community is focused on making AI systems more explainable to end users, a parallel, underserved population exists in the Global South. These individuals, often lacking basic literacy, are nevertheless engaged with various AI-driven products in their daily lives. These works so far has been centered around global north and very little is known about how XAI techniques have been designed, deployed, or tested with communities in the Global South. On the other side few works that focused on XAI in global South, most of the work was conducted in India, with less focus on other South Asian regions, Latin America, or Africa, and no work focused on Bangladesh. Only 3 out of 16 reviewed papers engaged with stakeholders or end users in their work [4, 14, 20]. Nearly all papers in the survey used explainability methods such as LIME or SHAP to explain how the features of the dataset contribute to the prediction by the model. LIME is a model-agnostic method that explains complex "black-box" model predictions by building a simple model around each prediction to identify key factors to explain individual explanations [24]. SHAP measures each feature's contribution to a model's prediction using game theory [16]. However, these methods require significant technical knowledge to understand and interpret them and might be difficult for local users. In a research study it has been seen that AI researchers and practitioners working on social impact problems and engaging with users in the Global South, participants mentioned while they

knew of XAI techniques, they lacked domain expertise, faced challenges in incorporating XAI to their workflow and perceived XAI as less useful for users with limited AI and digital literacy. The authors encouraged more work to understand what novice AI users need explained from AI systems to promote user agency [22]. Previous work suggests that in the absence of an explanation provided to the users, users often form incorrect mental models of how AI is working under the hood, which influences their decision-making [21].

Despite the huge number of XAI techniques, most of them are still challenging to comprehend due to the absence of human-centered design that meets human needs [32]. With literacy, there is another main reason for this: a lack of knowledge about AI and technology. Local users find technical stuff complex, so they simply like to ignore it. For this, even with technical explanations, users must be given relevant domain knowledge (e.g., what a feature signifies) and a broad awareness of how AI functions to attain practical understanding. Also, the "West" and "Global North" dominate AI design, development, and research [25]. Often, limited resources to develop region-specific AI tools make it difficult to produce their own AI systems, and these countries greatly depend on the products made by more technologically resourceful countries like the North. Bangladesh is a country full of diversity and different educational backgrounds. Instead of concentrating on how people use technologies only, XAI might consider what explainability, independence, and sovereignty (authority) mean to people from various backgrounds in this marginalized community. The same explanation that can satisfy a young businessperson, educator, or practitioner - a licensed individual who is legally allowed to offer services without being directly supervised might not be able to satisfy an older user with the same background. In the South Asian region, it is often found that female users are not very proficient with technology and face barriers such as lack of access and privacy [26, 30]. For this reason, they might need different explanations depending on the use cases.

3 TOWARDS EXPLAINABILITY IN BANGLADESH

One of the biggest existing challenges with AI systems in Bangladesh is that majority of the current systems are created with little or no explainability, highlighting the power dynamics between system developers and underprivileged groups who are using these systems. XAI techniques have been used in many AI-related research in Bangladesh in novel setting, such as using SHAP values to explain breast cancer classification [11], flood prediction [17] and student performance prediction [18]. Both SHAP and LIME techniques have been used to predict solar irradiation [27] and to model anxiety level of students in Bangladesh [10]. However, these works are done in research setting and have not been deployed to end users. As the following example demonstrates, end users in many industries, including healthcare and agriculture, will profit from the usage of explainability in AI systems.

Bangladesh's economy mostly depends on agriculture, and there has been a lot of advancement happening in recent times in this sector, and now it is in the era of using digital agriculture tools. A pilot study has been conducted on over 260 farmers from seven

districts of Bangladesh to understand the outcomes of these farming tools introduced beforehand [31]. From the study, it has been seen that female farmers are less likely to use these tools than men because of low education levels. Their knowledge of the tool's functionality was also constrained, which made them less interested or no interest in using these tools. So there is a big question of how can we create additional interventions that take into account the needs of women and eventually persuade them to use these tools. An app is in use that allows farmers to detect crop disease with a simple picture with computer vision techniques, but there is no explanation given to the users about the process. Some of the farmers said, in terms of full trust, they are very curious to learn about the processes of this magical output from the images. Furthermore, no locals have developed these tools, and nobody is available to respond to their inquiries. Their lack of clarity makes it difficult to convince the farmers to use them. They might be interested in employing these tools, as they can identify the percentage of the diagnosis that was likely caused by natural reason and the percentage that was caused by improper management if the model had been better created for explainability.

In a prior work, XAI was used to aid community health workers in rural India with visual and written explanations which helped to develop a more nuanced understanding of AI based predictions and improved their occupational abilities. Researchers recommended interactive visual heuristics and written explanations to promote active learning from the health workers who were the end users with limited digital literacy to operate an AI driven child malnutrition prediction tool [29].

Hence, we suggest that the human-centered XAI study should start with people rather than technology as what is perceived as reliable or useful by AI experts may differ greatly from what is considered reliable or useful by novice end-users. Giving human needs top priority when developing new technologies is also discussed and recommended in the work of Christine P Lee [15]. While designing mobile financial service (MFS) applications for the low-literate, marginalized communities in Bangladesh, authors found that these users face unique challenges, such as not understanding the meaning of common icons and forgetting the steps while operating specific applications. Some users derived the interpretation of icons from their cultural context, for example matching the settings icon with a tractor wheel and the music icon with a spoon, thus the researchers recommended the usage of culturally relevant icons and streamlining the workflow of the applications for the low literate user base [23]. For this demographic even if we use common XAI techniques such as provision of graphs for explainability, we need to simplify and contextualize them to obtain user centered explanations. One way could be to design tools with understandable language and additional features targeting users at various literacy levels. With language for the people who have no literacy, there could be added visuals, storytelling, voice based features and interactive dashboards. People who are tends to learn by observing or seeing how others are doing video demonstrations is a way to provide locals with an adaptive and self-explanatory explanation of the systems' outputs and usage. We can also consider other ideas for offline setting with limited and slow internet access in many parts of rural Bangladesh, such as sending sms for queries on specific predictions. Using analogies such as utilizing

tangible and familiar objects or processes to explain AI concepts may help the unlettered user demographics. Doing user research, particularly using qualitative techniques early on (such as creating personas, task models, and stakeholder engagement), can be another useful way to enhance acceptance. XAI researchers should identify certain intended user groups and design XAI solutions to fulfil their requirements. Individual users, including technical specialists, non-experts, and AI professionals, should use the XAI system in certain situations with specific requirements. While the recommendations in the paper are specific to the concerns of the users in Bangladesh, these considerations will also prove beneficial to other South Asian nations that share comparable demographic characteristics, as discussed in the work of Bloom DE et al. [2].

To know about the user needs specifically in this region, we plan to conduct a mixed-method study. With an interactive strategy, we hope to identify the challenges minorities face in utilizing explainable AI and work with them to co-design the concept of explainability in relation to dealing with consequential AI systems. The XAI system will be modified, and a framework will be developed based on the results of this formative research to offer the target participants a meaningful answer. Later on, we will evaluate the developed framework with the same participants to assess the system's potential.

4 CONCLUSION

With growing AI tools and decision models, proper user-centric explanations are needed for wider acceptance and to address responsible AI practice. In this work, we contend that further work is required to comprehend XAI system users, particularly using improved user requirements, to increase its practical effectiveness. While developing the systems, marginalized communities' needs should also be considered, and explainability should come from the concept of one-size-fits-all.

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